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Numerical Modelling of Acid Fracturing Across Fracture-Propagation Regimes

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Summary

Acid fracturing in tight formations exemplifies reactive flow coupled with rock deformation and crack growth, and remains a widely used stimulation technique. Predictive modeling is challenging because pressure-dependent apertures, tip-driven propagation, and wall dissolution interact across distinct propagation regimes. In this paper, we develop an integrated hydro-mechanical-chemical framework that combines a generalized 3D displacement discontinuity method (3D DDM) for the reversible mechanical aperture, an edge-cell finite-volume scheme for in-fracture flow and acid transport, reaction-controlled dissolution for the irreversible chemical aperture, and an LEFM-based, regime-dependent propagation strategy. Verification against analytical solutions for a penny-shaped fracture demonstrates excellent agreement in both toughness- and viscosity-dominated regimes. Application examples under viscosity domination reveal that the acid-rock reaction rate strongly governs etching patterns: a larger rate localizes dissolution near the inlet and shortens penetration, whereas a smaller rate extends the etched zone but promotes acid accumulation behind the tip. After shut-in, as the mechanical aperture elastically recloses, the irreversible chemical aperture governs residual conductivity. The approach is computationally efficient and geometrically robust for intersecting, non-coplanar fracture networks, providing a practical basis for optimizing acid-fracture design.

Key Words: Acid fracturing, 3D Displacement Discontinuous Method, Regime-dependent fracture propagation, Numerical modelling

Introduction

Hydraulic fracturing and acid fracturing are cornerstone stimulation techniques for low-permeability and carbonate formations, where limited matrix permeability and near-wellbore damage constrain productivity. In acid fracturing, an acid solution is injected to open and extend fractures while chemically etching their walls, thereby improving post-treatment conductivity (Zhu et al., 2024; Zeng et al., 2025). As the reactive fluid interacts with mineral constituents, it alters pore structure and local mechanical properties, which in turn affects fracture opening and propagation. Despite decades of practice, accurately predicting acid fracturing behavior under varying acid concentrations, injection rates, and acid-rock reaction rate remains challenging (Zeng et al., 2024a). Therefore, conducting simulations of acid fracture propagation holds significant value.

From the experimental and field sides, a growing body of work has clarified how acid type/strength, temperature, and injection schedule govern etching and post-treatment conductivity in carbonates, while emphasizing the need to couple chemistry with mechanics in design (Aljawad et al., 2019). True-triaxial laboratory tests in limestone show that injection rate and acid properties markedly affect pressure response and etched width, highlighting the sensitivity of fracture development to thermo-chemical conditions (Hou et al., 2019). Complementary experiments document that acid exposure alters pore structure and degrades mechanical properties of carbonate rocks, with implications for aperture evolution during and after treatment (Zhang et al., 2020). Field-oriented modeling and case evidence in naturally fractured carbonates further indicate that acid leak-off into intersecting natural fractures can curtail main-fracture growth even as it enhances conductivity within the natural-fracture network-an effect that must be accounted for in design and diagnosis (Ugursal et al., 2019). Integrated fracturing–reservoir workflows have since been used to generate spatial conductivity profiles and assess productivity under realistic conditions, enabling practical optimization of treatment parameters.

Fluid-driven fracture propagation couples rock deformation, fluid flow, and crack initiation/advance, with additional multiscale, multiphysics effects (Adachi et al., 2007). Numerical methods play a key role in understanding and predicting the complex processes involved in hydraulic fracturing propagation, and many methods have been proposed to address these challenges (Li et al., 2007). Continuum formulations such as finite elements (FEM/XFEM) capture complex geometries and stress states; classical FEM has long been used for viscosity-dominated hydraulic fracturing, while XFEM enrichments avoid remeshing and admit arbitrary crack paths (Fu et al., 2013; Gao and Ghassemi, 2020). Beyond these, phase-field approaches introduce a diffused damage variable that automates initiation, branching, and coalescence; recent developments have extended this paradigm to hydro-mechano-reactive acid fracturing systems (Zeng et al., 2024a; 2024b). In addition to continuum-based approaches, the discrete element method (DEM) has been used to simulate hydraulic fracture propagation (Yang et al., 2024). Moreover, the numerical manifold method (NMM) has been applied to model progressive rock-slope failure using a strength-based criterion, enabling the simulation of tensile, tensile-shear, and compressive-shear cracking (Yang et al., 2020). Beyond the above approaches, the displacement discontinuity method (DDM) has also been employed to analyze hydraulic-fracture propagation by focusing on the fracture boundary and its interaction with the surrounding formation. Because DDM incorporates analytical fundamental solutions and requires discretization only of the fracture surfaces, it achieves high accuracy at reduced computational cost, making simulations over large spatial domains tractable. Consequently, DDM has been widely adopted for hydraulic-fracturing problems in unconventional reservoirs and enhanced geothermal systems, with fracture-propagation models developed in 2-D (Cheng et al., 2020), planar 3-D (Zhang et al., 2017), nonplanar 2.5-D (Kresse and Weng, 2018; Chen et al., 2020), and fully nonplanar 3-D (Tang et al., 2019a; Fan et al., 2020) settings. However, to the best of our knowledge, DDM-based modeling of acid-fracture propagation remains limited: only a few studies have reported preliminary planar 3-D analyses using rectangular elements (Li et al., 2020; He et al., 2020). This gap motivates a fully three-dimensional, DDM-centered framework for acid fracturing, which can deepen understanding of acid-driven propagation and broaden the application scope of DDM to reactive fracturing problems.

In this work, we develop a fully three-dimensional, hydro-mechanical-chemical framework for acid fracturing that is centered on a generalized 3D-DDM for the reversible mechanical aperture, coupled to an edge-cell finite-volume scheme for in-fracture flow and acid transport, and a reaction-controlled dissolution model for the irreversible chemical aperture. Fracture propagation follows linear elastic fracture mechanics (LEFM) using tip-displacement stress-intensity factors, a maximum-principal-stress criterion (including twist) for direction, and a Paris-type law for quasi-static advance; a regime-based solution strategy is adopted with dynamic front tracking and Savitzky-Golay smoothing to maintain geometric robustness. The methodology is verified against analytical solutions for penny-shaped fractures in both regimes, and then applied to viscosity-dominated cases (15 wt% HCl) to examine how the acid-rock reaction rate governs concentration, pressure, and aperture fields as well as propagation. The remainder of the paper details the governing equations and numerical implementation, presents verification and application results, and discusses implications for optimizing the pairing of reaction rate and injection schedule in acid-fracture design.

Methodology

Building on our previous work (Fan et al., 2024), we employ a generalized 3D-DDM capable of simulating not only fracture problems but also boundary-value problems with non-symmetric closed contours. In the acid-fracturing numerical model developed here, the generalized 3D-DDM is used to compute the mechanical response to pressure within the fracture—expressed as the mechanical aperture—which is reversible. Fracture-internal fluid flow is solved using an edge-cell discretization FVM (Fan et al., 2025) to obtain the pressure field and the acid concentration field; these are then used to evaluate the chemically induced dissolution aperture, which is irreversible. During injection, once the propagation criterion is met at the fracture tip, the front is advanced by dynamically appending mesh elements so that the new crack front is tracked in real time until the prescribed simulation time is reached. The model relies on the following assumptions:

1. The fracture propagates in a linearly elastic medium with homogeneous Young's modulus and Poisson's ratio.
2. The injected fluid is an incompressible Newtonian fluid, and the flow within the fracture is governed by lubrication theory.
3. Only flow inside the fracture is considered; fluid flow in the surrounding matrix and poroelastic effects are neglected.
4. Dissolution at the fracture walls is reaction-controlled; i.e., the acid solute readily reaches the rock surface and mass-transfer limitations are negligible.
5. The fluid-lag zone between the crack front and the fluid front is negligible owing to the high confining stress in subsurface reservoirs (Gordeliy and Detournay, 2011).
6. Fracture growth is quasi-static and governed by the failure criterion of LEFM.

Fracture mechanical deformation

Building on our recent developments (Fan et al., 2022; 2024), we introduced a generalized 3D-DDM by improving the analytical expressions of the kernel integrals. This extension enables accurate and efficient computation of fracture aperture for complex fracture networks and fracture–cavity systems. In this model, each fracture is discretized into flat triangular elements (Fig. 1a). Every element is endowed with a local right-handed coordinate frame (Fig. 1b): the x-axis is aligned with a selected edge of the triangle (parallel to the x_1 vector; the z-axis is the unit normal x_2 to the element; and the y-axis is defined as $\boldsymbol{y} = \boldsymbol{x_2} \times \boldsymbol{x_1}$ to complete the triad. Within DDM, the displacement discontinuities across the fracture surfaces in the element-local coordinates are:

$$\begin{aligned} D_x &= u_x(x, y, 0_-) - u_x(x, y, 0_+) \\ D_y &= u_y(x, y, 0_-) - u_y(x, y, 0_+) \\ D_z &= u_z(x, y, 0_-) - u_z(x, y, 0_+) \end{aligned} \tag{1}$$

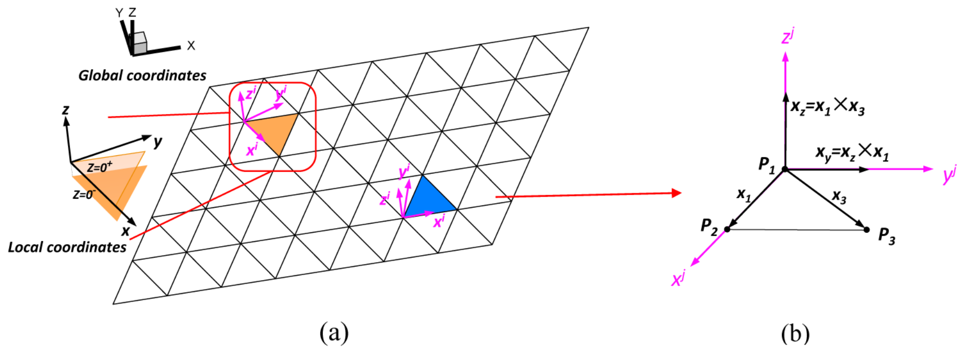

Figure 1—Illustration of global and local coordinates of 3D-DDM based on triangular elements: (a) two elements in the global coordinate system; and (b) setup of the element's local coordinate system (adapted from Fan et al. (2022).

where subscripts "0_+" and "0_-" denote the positive and negative sides of the fracture, respectively. The normal discontinuity equals the fracture aperture, whereas the two tangential discontinuities represent strike-slip and dip-slip shear. The governing relation between tractions and displacement discontinuities on element i is:

$$\begin{bmatrix}\tau_{s,f}^{i}\\ \tau_{t,f}^{i}\\ \sigma_{n,f}^{i}\end{bmatrix} = \sum_{N}^{j=1}\begin{bmatrix}C_{ss}^{ij} & C_{st}^{ij} & C_{sn}^{ij}\\ C_{ts}^{ij} & C_{tt}^{ij} & C_{tn}^{ij}\\ C_{ns}^{ij} & C_{nt}^{ij} & C_{nn}^{ij}\end{bmatrix}\begin{bmatrix}D_{s}^{j}\\ D_{t}^{j}\\ D_{n}^{j}\end{bmatrix} + \begin{bmatrix}\tau_{s,in}^{i}\\ \tau_{t,in}^{i}\\ \sigma_{n,in}^{i}\end{bmatrix} \tag{2}$$

where C is the influence coefficient of DDM; σ and τ are the normal and shear stresses; subscripts s, t, and n indicate strike-slip shear, dip-slip shear, and normal opening directions, respectively; *"f"* denotes the internal boundary stress due to fluid pressure, and *"in"* denote *in-situ* stresses. Indices i and j enumerate fracture elements, and N is the total number of elements. Thus, C_{ns}^{ij} is the influence coefficient that links the strike-slip shear displacement at element j to the normal displacement at element i. The influence coefficients are constructed from first-, second-, and third-order partial derivatives of the kernel function $\boldsymbol{I}(x,y,z)$:

$$\boldsymbol{I}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}) = \iint_{\Delta}\left[(\boldsymbol{x}-\boldsymbol{\xi})^{2}+(\boldsymbol{y}-\boldsymbol{\eta})^{2}+z^{2}\right]^{-1/2} \boldsymbol{d\xi d\eta} \tag{3}$$

which is the area integral of the Green's function over the triangular element expressed in the element-local coordinates. In the generalized 3D-DDM, we corrected and re-derived closed-form analytical expressions for $\boldsymbol{I}$ on triangular elements and its derivatives, achieving a typical 20–25 speed-up over numerical quadrature while maintaining accuracy. Consequently, Eq. 2 provides the instantaneous mechanical response (aperture) of the fracture to a given fluid-pressure field; i.e., any change in pressure leads to an immediate change in the mechanical aperture.

Fracture fluid flow

In this model, fractures are deformable. The mass balance in a fracture reads:

$$\frac{\partial\left(\rho_{f}\ \ w_{f}\right)}{\partial t} + \nabla\cdot\left(\rho_{f} w\, v_{f}\right) = Q_{sf} - Q_{leak} \tag{4}$$

where ρ_f is the fluid density; w_f is the (total) fracture aperture; Q_{leak} and Q_{sf} are the leak-off and source terms, respectively; and $\boldsymbol{v}_f$ is the in-fracture fluid velocity. Based on the cubic law, the velocity is:

$$\boldsymbol{v}_{f} = -\frac{w_{f}^{2}}{12\mu}\left(\nabla p - \rho_{f}\boldsymbol{g}\right) \tag{5}$$

with dynamic viscosity μ, pressure p, and gravity vector $\boldsymbol{g}$. In the present study, we neglect the gravitational contribution in Eq. 5 and assume no leak-off, i.e., $Q_{leak} = 0$. Applying the divergence theorem, the integral form of the mass balance over a control surface S with boundary ∂S becomes:

$$\int_s \rho_f \frac{\partial w_f}{\partial t} dS - \int_s Q_{sf} dS = \oint_{\partial S} \rho_f \frac{w_f^3}{12\mu} \frac{\partial p}{\partial n} dl \tag{6}$$

Because complex hydraulic-fracture networks contain numerous intersections and non-coplanar neighboring elements, we discretize Eq. 6 using an edge-cell FVM, which avoids spurious deviations in the flow direction from a control element to its non-planar neighbors. As illustrated in Fig. 2 for two intersecting non-planar elements (pink Cell-0 and green Cell-2), the edge-cell scheme first interpolates the pressure at the midpoint of a common edge from the cell-center pressures. The net flux from the control element (Cell-0) through its i-th common edge of length l_i is then computed as:

$$\frac{w_f^3}{12\mu} \frac{\partial p}{\partial \boldsymbol{n}} l_i = \frac{K_{0i}\, l_i}{12\mu\left(K_{0i} + \sum_{j=1}^{n} K_{ji}\right)} \left[\sum_{j=1}^{n} \left(K_{ji} P_j\right) - \left(\sum_{j=1}^{n} K_{ji}\right) P_0 \right] \tag{7}$$

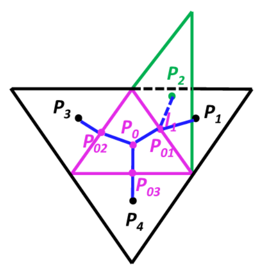

Figure 2—Illustration of edge-cell discretization FVM forms of fluid flow in non-coplanar elements (adapted from Fan et al. (2025).

with transmissibility-like coefficients

$$K_{0i} = \frac{\left(w_0\right)^3}{\boldsymbol{P_{0i}P_0} \cdot \boldsymbol{n}_{i0}}, \qquad K_{ji} = \frac{\left(w_j\right)^3}{\boldsymbol{P_{0i}P_j} \cdot \boldsymbol{n_{ij}}} \tag{8}$$

Here the subscript i indexes the three edges of the control element, and j indexes the neighboring elements sharing the i-th edge. P_0 and P_j are the pressures at the centers of Cell-0 and Cell-j, respectively; $\boldsymbol{P}_{0i}\,\boldsymbol{P_0}$ and $\boldsymbol{P_{0i}}$ $\mathbf{P}_{\boldsymbol{j}}$ are vectors from the midpoint of the i-th common edge to the corresponding cell centers; $\boldsymbol{n_{i0}}$ and $\boldsymbol{n_{ij}}$ are the outward normals from the same midpoint toward Cell-0 and Cell-j, respectively. As leak-off is neglected, the fracture system also satisfies an overall fluid volume balance:

$$\sum_{N}^{i=0} \int_{S_i} w_i dS_i = \int_0^t q_{in} dt \tag{9}$$

where S_i is the area of element i; w_i is the aperture; and q_{in} is the prescribed injection rate.

Fracture acid transport

During acid fracturing, transport of the acid species involves accumulation, advection, dispersion, and consumption by acid–rock reaction. The consumption term depends on the dominant mechanism—mass-transfer control (Panga et al., 2005; Jia et al., 2024), reaction control (Aljawad et al., 2020; Li et al., 2020), or mixed control (Zeng et al., 2024a; 2024b). In this work, we assume that the acid readily reaches the fracture walls so that the wall concentration equals the in-fracture concentration; the reaction is therefore reaction-controlled. The continuity equation for the acid species is:

$$\frac{\partial\left(w_f \quad \overline{C}\right)}{\partial t}+\nabla\cdot\left(w_f \quad \overline{C} \quad v_f\right)=\nabla\cdot\left(w_f \boldsymbol{D_A}\cdot\nabla\overline{C}\right)-2k_r\overline{C}^n+Q_{\overline{C}} \tag{10}$$

where $\overline{C}$ is the area-averaged acid concentration in the fluid phase, $\boldsymbol{D_A}$ is the effective dispersion tensor, k_r is the acid-rock reaction rate, n is the reaction order, and $Q_{\overline{C}}$ is an acid injection source term. For simplicity, k_r is treated as a constant. For completeness, under mass-transfer control the acid flux to the wall can be evaluated using the theoretical framework of Balakotaiah and West (2002). Combining Eq. 5 for the velocity (with gravity neglected) and applying the divergence theorem over a control surface S, the integral form of the acid transport equation reads:

$$\int_s \frac{\partial\left(w_f\overline{C}\right)}{\partial t}dS-\oint_{\partial S} w_f\boldsymbol{D_A}\nabla\overline{C}\cdot\boldsymbol{n}dl=\oint_{\partial S}\frac{w_f^3}{12\mu}\frac{\partial p}{\partial \boldsymbol{n}}\overline{C}dl-\int_s 2k_r\overline{C}^n dS+\int_s Q_{\overline{C}}dS \tag{11}$$

Consistent with the fluid-flow discretization, we employ the edge-cell FVM to discretize Eq. 11. For the advective term across a common edge between Cell-i and Cell-j, an upwind selection of concentration is used based on the sign of the edge-normal pressure gradient (with the same normal convention as in the flow discretization):

$$\overline{C}_{ij}=\begin{cases}\overline{C}_j & \left(\frac{\partial p}{\partial n}\right)_{ij}<0\\ \overline{C}_i & \left(\frac{\partial p}{\partial n}\right)_{ij}>0\end{cases} \tag{12}$$

This choice is then inserted into the edge flux evaluation in Eq. 11 to compute the advective transport consistently on intersecting, non-coplanar elements.

Fracture chemical deformation

Once the in-fracture acid concentration field is obtained, the dissolution-induced (chemical) aperture growth can be evaluated. The reaction-rate expression (Dong et al., 2002) is

$$R=\frac{2k_r\left(C_s\right)^n}{\rho_r}\cdot M_r \tag{13}$$

where R is the volume of rock dissolved per unit area and unit time (i.e., an equivalent recession velocity), ρ_r is the rock density,M_r is the rock molar mass, and C_s is the acid concentration at the rock surface, which equals $\overline{C}$ under the reaction-controlled assumption. Let v_a denote the acid–mineral stoichiometric coefficient (moles of acid required per mole of mineral). The rate of increase of chemical aperture follows:

$$\frac{\partial w_{chem}}{\partial t}=\frac{2k_r\left(\overline{C}\right)^n\cdot M_r}{v_a \quad \rho_r} \tag{14}$$

Accordingly, the total fracture aperture employed in the flow and transport modules is:

$$w_f\equiv w_{total}=D_n+w_{chem} \tag{15}$$

where D_n is the reversible mechanical aperture obtained from the generalized 3D-DDM, and w_{chem} is the irreversible dissolution aperture. Consequently, after shut-in, D_n decreases as the fracture progressively closes, whereas w_{chem} remains unchanged; acid fracturing therefore yields a significantly larger residual aperture than hydraulic fracturing, enhancing post-treatment productivity.

Fracture propagation

We assume fracture growth follows LEFM: a crack advances when the relevant stress intensity factor (SIF) reaches the material toughness, $\boldsymbol{K_i = K_{IC}}$. SIFs quantify the amplitude of the near-tip stress singularity; the three basic factors $\boldsymbol{K_I}$,$\boldsymbol{K_{II}}$, and $\boldsymbol{K_{III}}$ correspond to fracture modes I–III, respectively(Irwin, 1957). Common approaches to evaluate SIFs include the tip-displacement method (Lawn and Wilshaw, 1975), the ***J***-integral (Rice, 1968), the ***M***-integral (Moran and Shih, 1987), and domain equivalent integrals (Nikishkov and Atluri, 1987). Among these, the tip-displacement method is especially straightforward for DDM because it links SIFs directly to the opening and shear displacement discontinuities on the tip element. Following LEFM and previous studies (Olson, 2007; Wu and Olson, 2016; Li and Zhang, 2021), we compute:

$$\begin{aligned} K_I &= \xi \frac{D_n E\sqrt{\pi}}{4(1-v2)\sqrt{2r}} \\ K_{II} &= \xi \frac{D_s E\sqrt{\pi}}{4(1-v2)\sqrt{2r}} \\ K_{III} &= \xi \frac{D_t E\sqrt{\pi}}{4(1-v2)\sqrt{2r}} \end{aligned} \tag{16}$$

where r is the distance from the crack tip to the evaluation point (e.g., the tip-element centroid), and $\xi = 0.806$ is an empirical factor (Olson and Dahi-Taleghani, 2009). The fracture growth direction is determined using the maximum principal stress criterion of Schöllmann et al. (2002), which allows prediction of both deflection (kink) and twisting, thereby accommodating mixed-mode I/II/III propagation. Given the local stress state ahead of the tip in a cylindrical (polar) coordinate system (r,φ,z) attached to the crack front, the maximum principal stress is:

$$\sigma_1 = \frac{\sigma_\varphi + \sigma_z}{2} + \frac{1}{2}\sqrt{\left(\sigma_\varphi - \sigma_z\right)^2 + 4\tau_{\varphi z}^2} \tag{17}$$

where σ_φ, σ_z, and $\tau_{\varphi z}$ are the local components obtained by superposing far-field stresses with tip-induced stresses. The latter depend on the SIFs and can be expressed as (Schöllmann et al. 2002):

$$\begin{aligned} \sigma_\varphi^t &= \frac{K_I}{\sqrt{2\pi r}}\left[3\cos\left(\frac{\varphi}{2}\right) + \cos\left(\frac{3\varphi}{2}\right)\right] - \frac{K_{II}}{\sqrt{2\pi r}}\left[3\sin\left(\frac{\varphi}{2}\right) + \sin\left(\frac{3\varphi}{2}\right)\right] \\ \sigma_z^t &= \frac{\sigma_\varphi^t}{2} + \frac{1}{2}\sqrt{\left(\sigma_\varphi^t\right)^2 + 4\left(\tau_{\varphi z}^t\right)^2} \\ \tau_{\varphi z}^t &= \frac{K_{III}}{\sqrt{2\pi r}}\cos\left(\frac{\varphi}{2}\right) \end{aligned} \tag{18}$$

with $\boldsymbol{r}$ the radial distance from the tip and $\boldsymbol{\varphi}$ the polar angle in the local plane. The deflection angle $\boldsymbol{\varphi_0}$ that maximizes $\boldsymbol{\sigma_1}$ satisfies:

$$\left.\frac{\partial \sigma_1}{\partial \varphi}\right|_{\varphi=\varphi_0} = 0, \qquad \left.\frac{\partial^2 \sigma_1}{\partial \varphi^2}\right|_{\varphi=\varphi_0} < 0 \tag{19}$$

When mode-III loading is significant, the associated twisting angle ψ_0 along the front is given as a function of $\boldsymbol{\varphi_0}$:

$$\psi_0 = \frac{1}{2}\arctan\left[\frac{2\tau_{\varphi z}(\varphi_0)}{\sigma_\varphi(\varphi_0) - \sigma_z(\varphi_0)}\right] \tag{20}$$

After the growth direction is determined, we use the equivalent stress intensity factor K_{eq} proposed by Schöllmann et al. (2002) for mixed-mode propagation:

$$K_{eq} = \frac{\cos\left(\frac{\varphi_0}{2}\right)}{2}\left\{K_I\cos^2\left(\frac{\varphi_0}{2}\right) - K_{II}\frac{3\sin(\varphi_0)}{2} + \sqrt{\left[K_I\cos^2\left(\frac{\varphi_0}{2}\right) - K_{II}\frac{3\sin(\varphi_0)}{2}\right]^2 + 4K_{III}{}^2}\right\} \tag{21}$$

Propagation occurs when $K_{eq} = K_{IC}$ Because many geomaterials (e.g., quartzite, sandstones, granites, marbles) exhibit subcritical (quasi-static, stable) crack growth below the critical SIF, we model the incremental advance using a Paris-type law widely adopted in the literature (Gupta and Duarte, 2014; Lazarus et al., 2001; Thomas et al., 2020; Li et al., 2020; Li and Zhang, 2023):

$$\Delta d = \Delta d_{max} \left(\frac{K_{eq}}{K_{IC}} \right)^{m} \tag{22}$$

Here Δd_{max} is the user-specified maximum advance per step, and m is a material exponent (set to 1 in this work). The fracture toughness K_{IC} is taken as constant during propagation.

For numerical robustness, all tips are advanced simultaneously at each step with possibly different magnitudes Δd^i and angles φ_0^i, ψ_0^i. This is appropriate when tip speeds are comparable within a time step (Tang et al., 2019a; Wong and Cui, 2023); otherwise, jagged fronts and distorted elements may arise, risking solver failure. To regularize the geometry, we smooth the newly generated front using a third-order Savitzky-Golay filter (Savitzky and Golay, 1964) to remove outliers and take the smoothed nodes as the new tip positions. For example, in Fig. 3a the red loop denotes the previous front, the blue loop the raw updated front, and the pink loop the smoothed front; pronounced zigzags are mitigated. Fig. 3b shows the elements appended to the previous front.

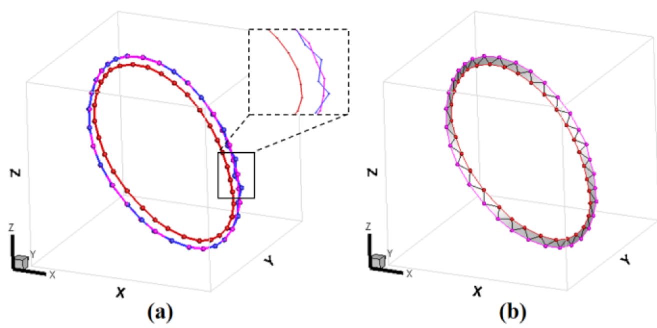

Figure 3—Smooth of the penny-shaped fracture front: (a) front before and after 3rd order SG filter; and (b) the final adding elements to the previous fracture front.

Regime-based solution strategy

Over the last two decades, substantial efforts have been devoted to clarifying propagation regimes in hydraulic fracturing (Bunger et al., 2005; Peirce and Detournay, 2008; Donstov, 2016; Lecampion et al., 2018). Based on the ratio between the energy dissipated in the rock to create new fracture surfaces and the viscous dissipation in the fluid, energy arguments (Garagash and Detournay, 2000; Savitski and Detournay, 2002) indicate three regimes of propagation: viscosity-dominated, toughness-dominated, and transient. For penny-shaped fractures, Savitski and Detournay (2002) and Zhang et al. (2002) introduced the following dimensionless groups via a similarity (nondimensional) analysis:

$$\begin{aligned} \boldsymbol{K} &= K' \left(\frac{t^2}{\mu'^5 Q_0^3 E'^{13}} \right)^{1/18} \\ \boldsymbol{M} &= \mu' \left(\frac{Q_0^3 E'^{13}}{K'^{18} t^2} \right)^{1/5} \end{aligned} \tag{23}$$

where E' is the plane strain modulus and μ' and K' are defined to simplify numeric factors:

$$\mu' = 12\mu, \quad E' = \frac{E}{1-\nu^2} \quad K' = 4\left(\frac{2}{\pi}\right)^{1/2} K_{Ic} \tag{24}$$

When $\boldsymbol{K} \ll 1$, the fracture is viscosity-dominated (toughness effects are negligible); the energy to create new surfaces is small relative to viscous dissipation. Conversely, when $\mathbf{K} \gg 3.5$, the fracture is toughness-dominated (viscous effects are negligible); viscous dissipation is small compared with the energy dissipated at the crack tip (Savitski and Detournay 2002). These regimes feature distinct pressure characteristics: large pressure gradients within the fracture in the viscosity-dominated regime versus nearly uniform pressure in the toughness-dominated regime. Accordingly, we adopt regime-specific-based numerical solution strategies.

Irrespective of regime, we employ a weak coupling between the flow/pressure–aperture system and the acid transport/chemistry: within each time step, the dissolution-induced aperture increment is applied once at the end of the step to update the pressure and mechanical aperture, without inner iterations. This is justified by the relatively small variations in pressure and mechanical response due to dissolution over a single step and leads to a substantial gain in efficiency.

Toughness-dominated regime

When propagation is toughness-dominated, the pressure field is approximately spatially uniform within the fracture (Tang et al., 2022; Tang et al., 2018). We therefore compute the time-step pressure and mechanical aperture directly from the fluid volume balance (Eq. 9) using a bisection search on the uniform pressure. After obtaining the end-of-step pressure and mechanical aperture, we solve for the acid concentration and the corresponding dissolution-aperture increment. Because the pressure is uniform, no further pressure/mechanical update is required; we simply superpose the mechanical and chemical apertures and test the propagation criterion. The numerical treatment of toughness-dominated propagation regime is briefly outlined as follows:

1. **Initialization:** set input parameters; set time step $k = 0$, step size Δt, and the start-of-step element state (mechanical aperture $w_m = D_n$, pressure p, concentration $\overline{C}$).
2. **Loop while** $t < t_{final}$:
   - Use Eq. 9 and bisection to find the uniform pressure p^k that satisfies volume balance; compute the corresponding mechanical aperture field w_m^k via the generalized 3D-DDM.
   - Using Eqs. 11 and 14, solve for the acid concentration field and obtain the dissolution increment Δw_{chem}^k; update the total aperture $w^k = w_m^k + \Delta w_{chem}^k$.
   - Check propagation: if the criterion is met, update the fracture front and update the influence matrix C; set $k \leftarrow k + 1$, $t \leftarrow t + \Delta t$, and continue.
   - Otherwise, adapt Δt according to the propagation status, restore the start-of-step state, and repeat from step 2.1.

Viscosity-dominated regime

When propagation is viscosity-dominated, pressure gradients are significant. In this case, the coupled flow–mechanics problem at each time step must be obtained through an iterative procedure, such as the Newton-Raphson method (Mao et al., 2023) or the Picard iteration (Tang et al., 2019b; Wu and Olson, 2015). Given the pressure and mechanical aperture, we then compute the acid concentration and the dissolution increment. Because pressure gradients are non-negligible, we perform one additional update of the pressure and mechanical aperture using the enlarged aperture (consistent with the weak-coupling strategy). The total aperture is then used to evaluate the propagation criterion. The numerical procedure of toughness-dominated propagation regime is summarized outlined as follows:

1. **Initialization:** set input parameters; set time step $k = 0$, step size Δt, and the start-of-step element state (mechanical aperture $w_m = \mathrm{D_n}$, pressure p, concentration $\overline{C}$).
2. **Loop while** $t < t_{final}$:

2.1 Using Eqs. 2 and 6, solve for the quasi-steady pressure p^k and mechanical aperture w_m^k by Picard iteration.

2.2 Using Eqs. 11 and 14, solve for the acid concentration and compute Δw_{chem}^k.

2.3 Single weak-coupling update: recompute p_k^{+1} and w_m^{k+1} once with the updated aperture $w^k = w_m^k + \Delta w_{chem}^k$.

2.4 Check propagation: if the criterion is met, update the fracture front and update the influence matrix C; set $k \leftarrow k + 1$, $t \leftarrow t + \Delta t$, and continue.

2.5 Otherwise, adapt Δt according to the propagation status, restore the start-of-step state, and repeat from step 2.1.

Because field observations indicate that most hydraulic/acid-fracturing treatments operate in the viscosity-dominated regime (Bunger and Detournay, 2008), our case studies focus on viscosity-dominated acid-fracture propagation.

Algorithm verification

This section validates the correctness and accuracy of the regime-based fracture-propagation algorithm using a single penny-shaped fracture for which an analytical solution is available. For the validation case, pure water is used as the injection fluid. The remaining parameters are summarized in Table 1 and, unless otherwise stated, are retained in all subsequent numerical studies.

Table 1—Physical parameter values used in the verification cases

Parameters	Unit	Value
Injection flow rate	$m^3 \cdot s^{-1}$	0.01
Fluid density	$kg \cdot m^{-3}$	1000
Young's modulus	GPa	38.8
Poisson's ratio	--	0.15
In-situ stress in *x* direction	MPa	0.0
In-situ stress in *y* direction	MPa	2.0
In-situ stress in *z* direction	MPa	0.0
Initial fracture aperture	mm	0.01

Toughness-dominated propagation benchmark

The schematic of the penny-shaped hydraulic fracture model is shown in Fig. 4a. Injection occurs at the fracture center, with an initial radius $R_0 = 6.5\ m$; the allowed advance per propagation step is $\Delta d_{max} = 3.0\ m$. In the toughness-dominated regime, the analytical solutions for inlet width, aperture, and fracture radius as functions of time are (Savitski and Detournay, 2002; Dontsov, 2016):

$$\begin{aligned} w(\rho, t) &= 0.6537*\left[\frac{K'^4 Q_0 t}{E'^4}\right]^{\frac{1}{5}}(1-\rho^2)^{\frac{1}{2}} \\ R(\rho, t) &= 0.8546*\left[\frac{E' Q_0 t}{K'}\right]^{\frac{2}{5}} \\ P(\rho, t) &= 0.3004*\left[\frac{K'^6}{E' Q_0 t}\right]^{\frac{1}{5}} \end{aligned} \tag{25}$$

where Q_0 is the injection volumetric rate and $\rho = r/R(t)$ is the dimensionless radial coordinate. From these expressions it follows that toughness-dominated propagation is independent of fluid viscosity.

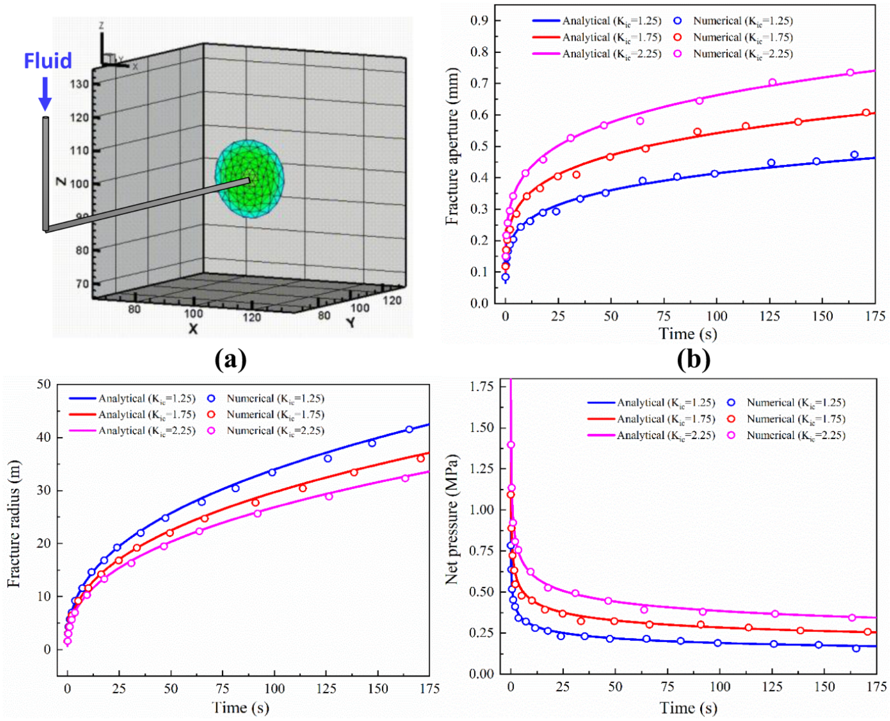

Figure 4—Verification of the toughness-dominated fracturing solution strategy using a single penny-shaped fracture: (a) schematic of the model; (b) fracture aperture comparison; (c) fracture radius comparison; and (d) fracture pressure comparison.

Figs. 4b–4d compare, for three fracture toughness values K_{IC} = 1.25, 1.75, 2.25, the injection-point aperture, fracture radius, and injection-point net pressure versus time; hollow circles mark the simulated values at each attainment of the critical propagation state. Excellent agreement is observed between the numerical results and the analytical solutions, validating the toughness-dominated, regime-based strategy. Moreover, increasing K_{IC} raises the energy required to create new surfaces; accordingly, the internal pressure is higher for larger toughness (Fig. 4d), which yields a larger inlet aperture (Fig. 4b) and a slower propagation rate (Fig. 4c).

To illustrate the dynamic evolution under toughness domination, Figs. 5a–5c plot the aperture fields at K_{IC} = 1.75 for timesteps 3, 8, and 13. The aperture at the fracture center increases as the fracture grows, while the tip-element aperture remains essentially unchanged because the fracture toughness is constant during the simulation. After each growth step, a third-order Savitzky-Golay filter is applied to smooth the newly generated front, yielding a regular tip shape without jagged elements. In addition, because the advance distance is set by a Paris-type law, tip speeds within a given time step are comparable across elements, leading to a fairly uniform set of newly added elements at each step.

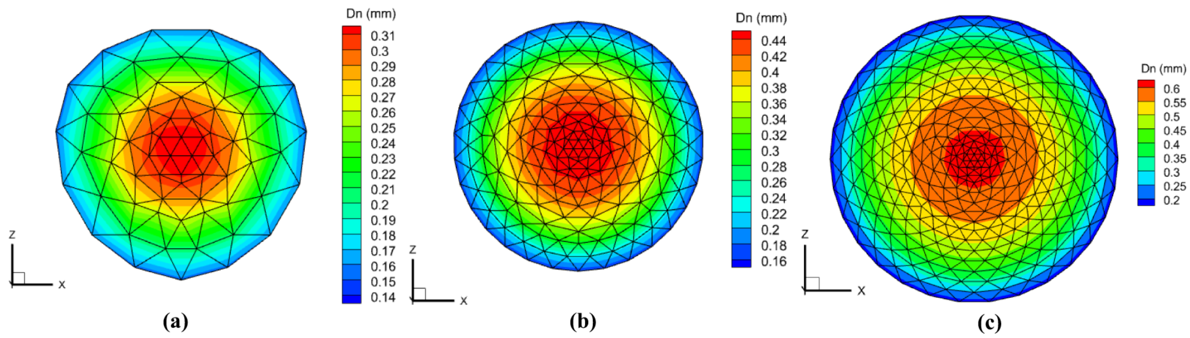

Figure 5—Fracture-front evolution and corresponding aperture distributions at different times (K_{IC} = 1.75): (a) Timestep = 3; (b) Timestep = 8; and (c) Timestep = 13.

Viscosity-dominated propagation benchmark

The same penny-shaped-fracture model as in Fig. 4a is used to verify the viscosity-dominated fracturing strategy, with the same initial radius R_0 = 6.5 m, and per-step advance Δd_{max} = 3.0 m. In the viscosity-dominated regime, the analytical solutions for inlet width, aperture, and radius are (Savitski and Detournay, 2002; Dontsov, 2016):

$$\begin{aligned} w(\rho, t) &= 1.1901 * \left[\frac{\mu'^2 Q_0^3 t}{E'^2}\right]^{\frac{1}{9}} (1+\rho)^{0.487} (1-\rho)^{\frac{2}{3}} \\ R(\rho, t) &= 0.6944 * \left[\frac{Q_0^3 E' t^4}{\mu'}\right]^{\frac{1}{9}} \\ P(\rho, t) &= \Pi_{m0} * \left[\frac{\mu' E'^2}{t}\right]^{\frac{1}{3}} \\ \Pi_{m0}^{(1)} &= A_1^{(1)}\left[\omega_1 - \frac{2}{3(1-\rho)^{1/3}}\right] - B^{(1)}\left[ln\frac{\rho}{2} + 1\right] \end{aligned} \tag{26}$$

where Π_{m0} = $\Pi_m(\rho, 0)$ is the zero-toughness dimensionless net pressure, sought via series expansions (Savitski and Detournay, 2002). Here we adopt its first-order approximation with $A_1^{(1)} = 0.3386$, $B^{(1)} = 0.0932$, $\omega_1 = 2.479$. As indicated by Eq. 26, viscosity-dominated propagation depends on fluid viscosity. Accordingly, we simulate cases with μ = 0.01, 0.1, 1.0 (while fixing K_{IC} = 1.25). Figs. 6a–6c show the time evolution of injection-point aperture, net pressure, and radius for different viscosities. The numerical solutions exhibit excellent agreement with the analytical results, confirming the correctness of the viscosity-dominated, regime-based strategy. Furthermore, with increasing viscosity, viscous dissipation requires more energy; thus the injection-point pressure is higher (Fig. 6b), more fluid accumulates near the inlet, the inlet aperture increases (Fig. 6a), and the propagation rate decreases (Fig. 6c).

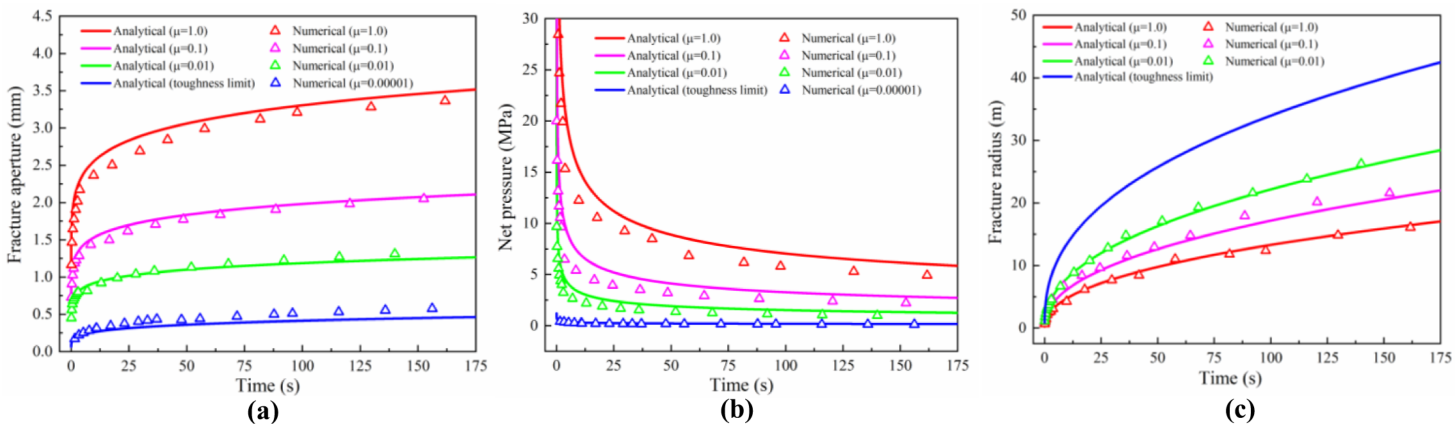

Figure 6—Verification of the viscosity-dominated fracturing solution strategy using a single penny-shaped fracture: (a) fracture aperture comparison; (b) fracture pressure comparison; and (c) fracture radius comparison

To visualize the dynamic evolution under viscosity domination, Figs. 7a–7c and 7d–7f plot, respectively, the aperture fields and the pressure fields at $K_{IC} = 1.25$ and $\mu = 1.0$ for timesteps 3, 8, and 12. As seen in Figs. 7d-7f, substantial pressure gradients develop within the fracture, necessitating an iterative solution for the fluid pressure and mechanical response. As the fracture grows, the inlet aperture increases while the inlet pressure decreases, owing to the continually expanding fracture volume. If the fluid viscosity continues to increase, the fluid front will no longer coincide with the crack front, leading to the emergence of a fluid-lag zone.

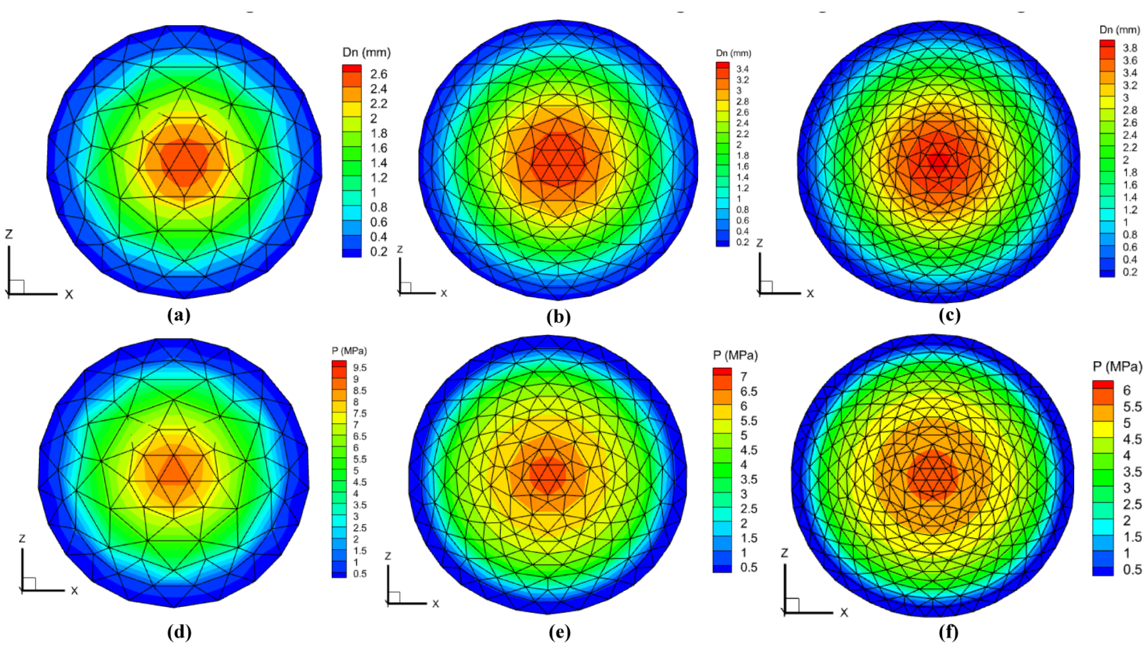

Figure 7—Fracture aperture distributions at different times (K_{IC} = 1.25, μ = 1.0): (a) Timestep = 3; (b) Timestep = 8; and (c) Timestep = 12; and the fracture pressure distributions at different times: (d) Timestep = 3; (e) Timestep = 8; and (f) Timestep = 12.

Application and discussion

In this part, we present preliminary numerical simulations of acid fracturing under viscosity-dominated conditions to examine the influence of different acid–rock reaction rates. Three cases are designed—Case 1-Case 3—with reaction rates $k_r = 10^{-6}$, 10^{-5}, 10^{-4}. We use the same penny-shaped fracture model as in Fig. 4a, with the initial radius $R_0 = 6.5m$ and the per-step advance $\Delta d_{max} = 1.5\ m$ A 15 wt% HCl solution is injected at the fracture center. The remaining parameters are summarized in Table 2.

Table 2—Physical parameter values used in the acid fracturing simulation

Parameters	Unit	Value
Injection flow rate	$m^3 \cdot s^{-1}$	0.01
Injection acid concentration	%	15
Young's modulus	GPa	38.8
Poisson's ratio	--	0.15
Acid density	$kg \cdot m^{-3}$	1045
Acid molecular weight	$g \cdot mol^{-1}$	36.46
Acid viscous	$Pa \cdot s^{-3}$	1.0
Rock density	$kg \cdot m^{-3}$	2650
Rock molar mass	$kg \cdot mol^{-1}$	0.1
In-situ stress in x direction	MPa	0.0
In-situ stress in y direction	MPa	2.0
In-situ stress in z direction	MPa	0.0
Initial fracture aperture	mm	0.01
Initial acid concentration	%	0.0

Impact on the fracture concentration field

Because dissolution at the fracture wall is reaction-controlled, the acid-rock reaction rate directly affects the extent of etching: a larger k_r implies faster acid consumption. Fig. 8 shows the concentration fields at timesteps 2, 5, and 8 for the three cases (panels **a1-a3**: Case 1; **b1-b3**: Case 2; **c1-c3**: Case 3), with the fracture at the critical propagation state at each timestep.

Comparing different cases at the same timestep (e.g., Fig. 8a1, 8b1, 8c1) shows that increasing k_r leads to greater acid consumption near the injection point and, consequently, a shorter penetration distance (i.e., a smaller effectively etched zone). When k_r is small, the effectively etched zone is longer, but continuous injection causes acid accumulation within the fracture fluid. As illustrated in Fig. 8a3, the relatively small k_r and the small tip aperture limit the available fluid volume near the front, producing elevated acid concentrations within a finite distance behind the tip—a trend that becomes more pronounced as injection proceeds. For intermediate k_r, a more balanced acidizing effect is observed (Fig. 8b1–b3). The choice of k_r should ultimately be considered jointly with the injection rate; detailed exploration of the optimal k_r-rate pairing is left for future work.

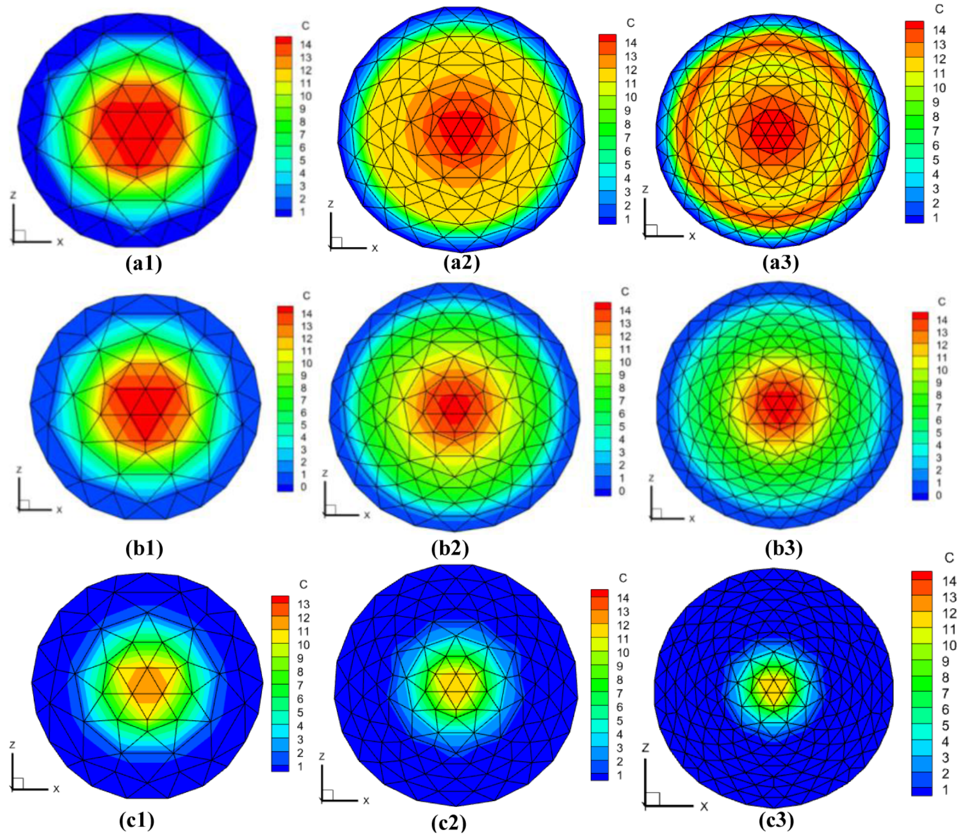

Figure 8—Fracture concentration field at timesteps 2, 5, and 8 for different cases: (a1-a3) Case 1; (b1-b3) Case 2; and (c1-c3) Case 3.

Impact on the fracture chemical-aperture field

Under the reaction-controlled assumption, the chemical (dissolution-induced) aperture is directly linked to the concentration field. Fig. 9 plots the cumulative chemical aperture at timesteps 2, 5, and 8 for the three cases (**a1-a3**: Case 1; **b1-b3**: Case 2; **c1-c3**: Case 3). Note that the contour scales differ across panels.

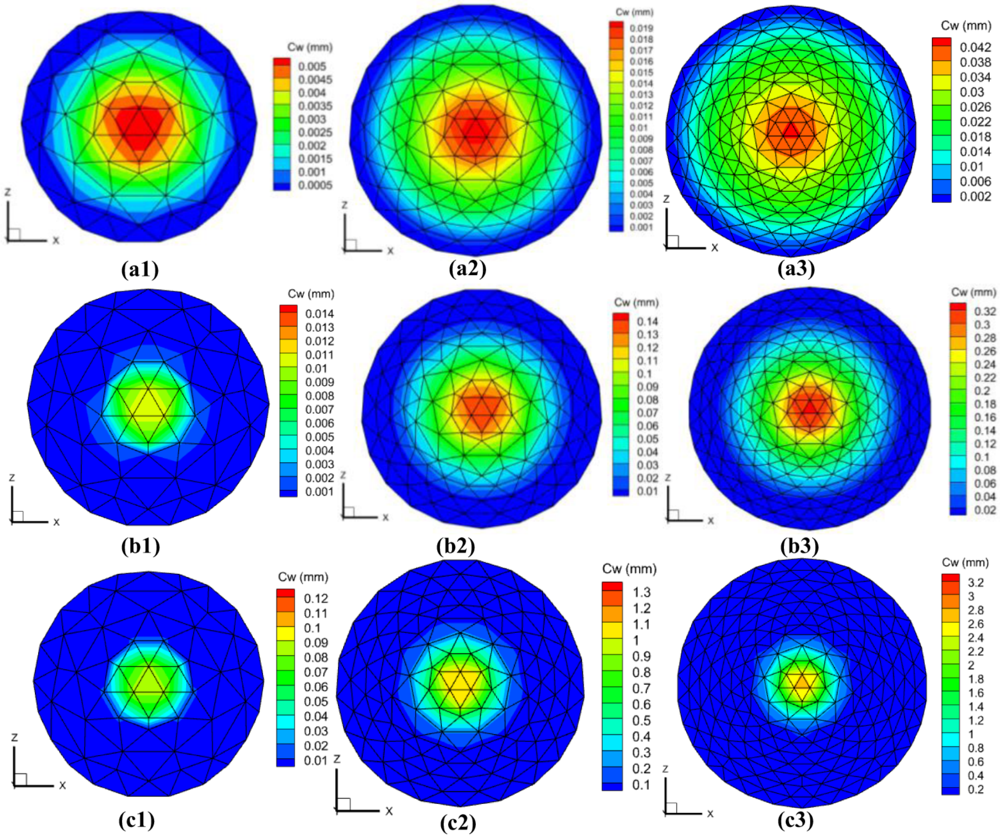

Figure 9—Fracture chemical-aperture field at timesteps 2, 5, and 8 for different cases: (a1-a3) Case 1; (b1-b3) Case 2; and (c1-c3) Case 3.

From Fig. 9a1, 9b1, and 9c1, the chemical aperture becomes increasingly localized around the injection point as k_r increases, mirroring the concentration patterns. In Case 3, the acid is exhausted within a very short distance from the inlet, and the chemical aperture there is substantially larger than in the other cases.

Impact on the fracture pressure field

The growth of the chemical aperture, in turn, affects the pressure distribution and thus the propagation process. Fig. 10 shows the pressure fields at timesteps 2, 5, and 8 (**a1-a3**: Case 1; **b1-b3**: Case 2; **c1-c3**: Case 3). From Figs. 10a1-a3 and Figs. 10b1-b3, the pressure fields in Case 1 and Case 2 are essentially similar up to timestep 8. Although k_r differs, the etched regions remain limited and the magnitudes of the chemical aperture are still small, so their influence on pressure is minor. In Case 3, the etched region occupies a smaller portion of the fracture but exhibits a larger cumulative chemical aperture, which already affects the pressure distribution (see Fig. 10c3). As time advances, the etched region in Case 2 is expected to cease expanding while the chemical aperture within it continues to accumulate, eventually producing a pressure response similar to that observed in Case 3.

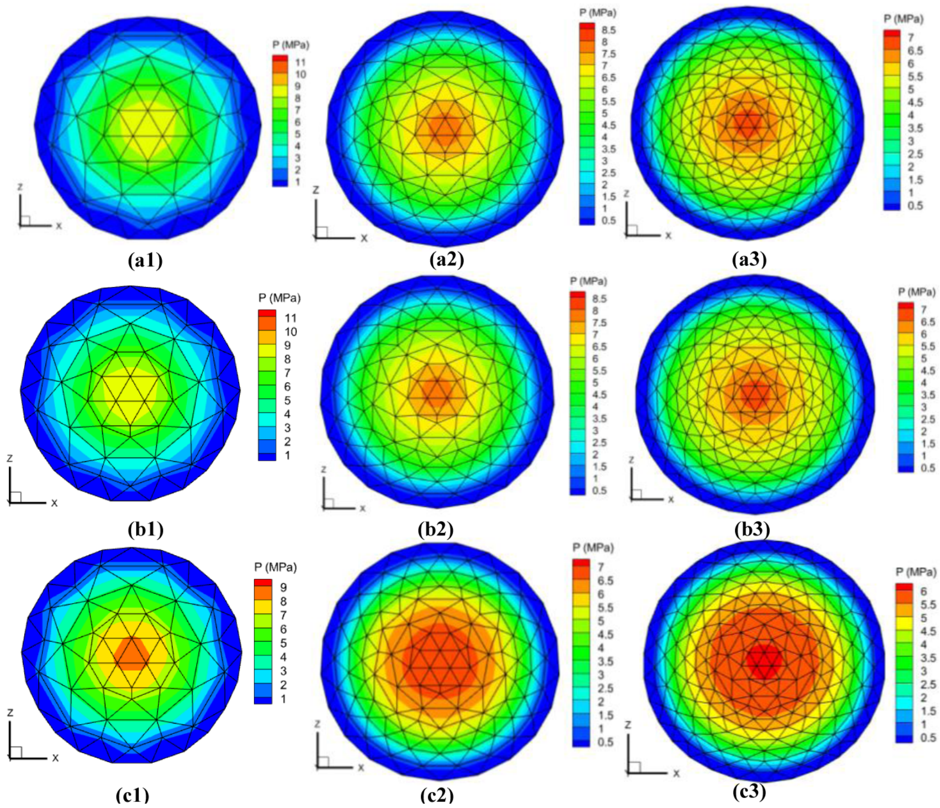

Figure 10—Fracture pressure field at timesteps 2, 5, and 8 for different cases: (a1-a3) Case 1; (b1-b3) Case 2; and (c1-c3) Case 3.

Impact on fracture mechanical aperture field

The mechanical aperture is the instantaneous elastic response to the pressure field. Fig. 11 presents the mechanical aperture at timesteps 2, 5, and 8 for the three cases (**a1-a3**: Case 1; **b1-b3**: Case 2; **c1-c3**: Case 3). Although the pressure field of Case 3 differs from those of Cases 1 and 2, the corresponding mechanical-aperture responses are not markedly different across cases. This is because the simulations cover only eight timesteps (≈130 s of injection): the time span is too short for the pressure variations to drive substantial differences in elastic opening, and the overall changes in fracture aperture remain modest.

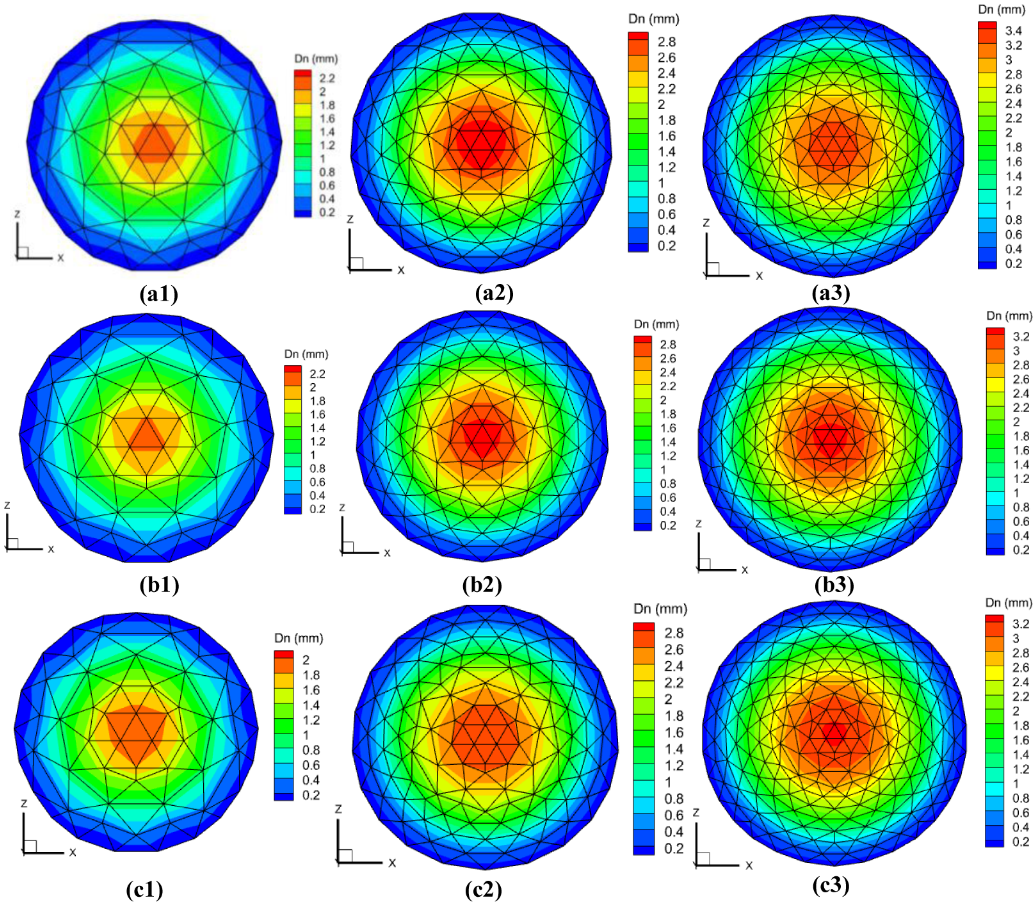

Figure 11—Fracture mechanical aperture field at timesteps 2, 5, and 8 for different cases: (a1-a3) Case 1; (b1-b3) Case 2; and (c1-c3) Case 3.

Conclusions

This work develops a coupled hydro-mechanical-chemical framework for acid fracturing that integrates a generalized 3D DDM, an edge-cell FVM for flow and transport, and an LEFM-based, regime-dependent propagation strategy. Validation against analytical solutions for a penny-shaped fracture confirms accuracy in both toughness- and viscosity-dominated regimes. Preliminary applications under viscosity domination examine the impact of acid-rock reaction rate. The main conclusions are summarized below:

1. The regime-based solution strategy reproduces analytical behavior: in the toughness-dominated regime, pressure is nearly uniform and independent of viscosity; higher fracture toughness increases inlet pressure and aperture while slowing radial growth.
2. In the viscosity-dominated regime, significant pressure gradients necessitate an iterative solve; higher viscosity raises inlet pressure, increases inlet aperture, and reduces propagation rate.
3. The acid–rock reaction rate k_r controls etching patterns: larger k_r localizes dissolution near the inlet and shortens penetration; smaller k_r extends the etched zone but promotes acid accumulation behind the tip; intermediate k_r yields more balanced acidizing.
4. Over the short simulation window considered (eight timesteps, ~130 s), differences in pressure produce only modest variations in mechanical aperture, whereas the chemical aperture grows irreversibly and thus governs residual opening.
5. The proposed framework is computationally efficient and geometrically robust (simultaneous tip advance with Savitzky-Golay smoothing), providing a practical tool for studying acid-fracture propagation and for future optimization of k_r-injection-rate combinations.

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